

# JAMES OPATA TETTEH

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## PROFILE

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Electrical Engineering graduate with practical utility and industrial experience from the Volta River Authority and Ghana Water Limited. Skilled in electrical system maintenance, motor diagnostics, project execution, and technical documentation with experience in data collection, analysis, and reporting. Possesses a strong technical foundation in SCADA systems, CAD design, and network simulation software. Proven ability to lead data collection teams in the field, troubleshoot control systems, and design scalable automated processes. Effective communicator and team player with the ability to work in both field and office environments. Passionate about sustainability, climate action, and contributing to multidisciplinary engineering and research-driven projects.

## EDUCATION

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### BSc. Electrical/Electronic Engineering

November, 2025

**Undergraduate Project:** LPG Leakage Detector for LPG-Driven Vehicle

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

### W.A.S.S.C.E (General Science)

September 2018 – October 2021

St Peter's Senior High School, Nkwatia-Kwahu, Ghana

## WORK EXPERIENCE

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### National Service Person – Production Machine Operator, Fairafric Ghana Limited | Nov 2025 – Present

- Operate and monitor automated chocolate production equipment to maintain optimal process efficiency, product quality, and production throughput.
- Perform routine preventive maintenance and first-line troubleshooting on industrial machinery including motors, sensors, and control systems to minimize downtime.
- Monitor key process parameters (temperature, speed, flow rates) to ensure process stability, quality assurance, and compliance with industrial safety standards, while supporting continuous process improvement initiatives.

### Electrical Engineering Intern, Volta River Authority, Akuse | September – November 2024

- Supported project implementation and stringent quality control for facility electrical installations.
- Assisted and in the technical supervision, wiring, and installation of automated boom barrier systems.
- Collaborated directly with senior utility engineers to guarantee adherence to rigorous safety and quality standards.

### Electrical Engineering Intern, Ghana Water Limited, Kpong | September – December 2023

- Supported diagnosis and repairs of electrical motors, soft starters, and control panels, ensuring operational compliance.
- Conducted detailed inspections and scheduled preventive maintenance of high-load electrical motors.
- Led a team of interns to successfully collect critical vibration and bearing temperature data using a vibrometer.

## PROJECTS/RESEARCH EXPERIENCE

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- Load-Flow Analyzer (Python): Developed a power flow calculator for electrical network simulation.
- LPG Monitoring and Leakage Detection System: Built an Arduino-based safety system for gas leakage alerts.
- Smart Fan System: Designed an automatic temperature-controlled fan incorporating sensor feedback loops for home applications.
- Direct-On-Line Starter: Implemented robust DOL design for a submersible pump system.
- Automate Juice Filtration System: Designed an automated juice filtration system for production processes utilizing PLC logic and P&ID frameworks to support continuous process improvement

## PROFESSIONAL TRAINING/ CERTIFICATION

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|----------------|------------------------------------------------------|-------|
| AI Starter Kit | 17 <sup>th</sup> March -19 <sup>th</sup> April, 2025 | ALX   |
| MySQL          | 7 <sup>th</sup> -10 <sup>th</sup> February, 2025     | Cursa |

## SKILLS AND ABILITIES

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- Technical: C, Python, MySQL, MATLAB, SCADA Systems, KiCad, Eagle CAD, Microsoft Office Suite
- Soft Skills: Teamwork, Analytical Thinking, Communication, Adaptability, Problem-Solving

## LANGUAGES

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- Fluent – English, Dangme
- Beginner – French, Twi

## REFEREES

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### Mr. Dominic Bampo

Project Manager, KNUST

Consultant, United State Agency for International Development (USAID) Real Water

CEO, Institute of Scientific Research and Consultancy

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### Ing. Luke Sitsofe Kwabla Dei

Principal Electrical Technician Engineer, Volta River Authority

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